

Activities and intents

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Activities and intents

- What activities are
- Understanding the concept of intents How to use the
- Events of an Activity
- Life Cycle of an Activity
- Intent object to link activities
- Explicit and Implicit Intents

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Understanding Activities

```
package net.learn2develop.Activities;
import android.app.Activity;
import android.os.Bundle;
public class MainActivity extends Activity {
/** Called when the activity is first created. */
@Override
public void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.main);
}
}
```

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Activities

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="net.learn2develop.Activities"
    android:versionCode="1"
    android:versionName="1.0">
    <application android:icon="@drawable/icon"
        android:label="@string/app_name">
        <activity android:name=".MainActivity"
            android:label="@string/app_name">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category
                    android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
    </application>
    <uses-sdk android:minSdkVersion="9" />
</manifest>
```

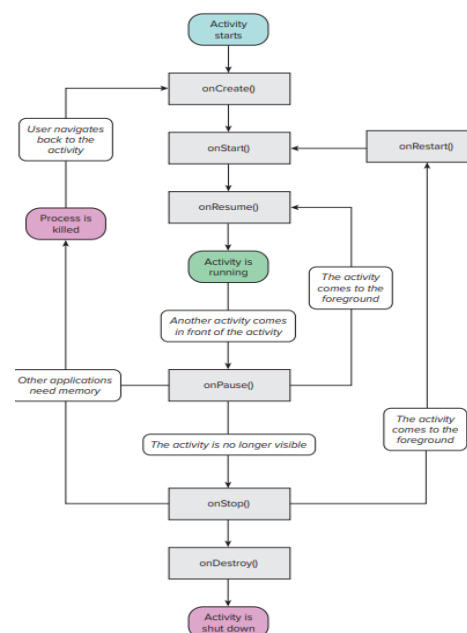
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Events of an Activity

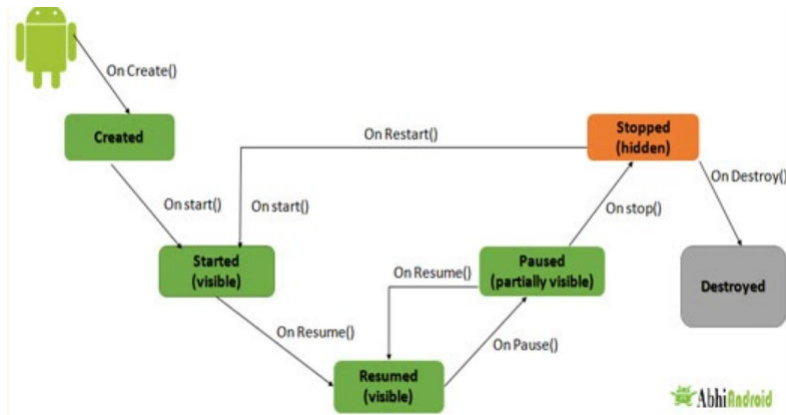
- `onCreate()` — Called when the activity is first created
- `onStart()` — Called when the activity becomes visible to the user
- `onResume()` — Called when the activity starts interacting with the user
- `onPause()` — Called when the current activity is being paused and the previous activity is being resumed
- `onStop()` — Called when the activity is no longer visible to the user
- `onDestroy()` — Called before the activity is destroyed by the system (either manually or by the system to conserve memory)
- `onRestart()` — Called when the activity has been stopped and is restarting again

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Life Cycle of an Activity



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LIST OF ACTIVITY LIFECYCLE METHODS OR STATES:

- **Activity Created: onCreate(Bundle savedInstanceState): onCreate()**
- **Activity Started: onStart():**
- **Activity Resumed: onResume():**
- **Activity Paused: onPause():**
- **Activity Stopped: onStop():**
- **Activity Restarted: onRestart():**
- **Activity Destroyed: onDestroy():**

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linking Activities using intents

- An Android application can contain zero or more activities.
- When your application has more than one activity, you may need to navigate from one activity to another.
- In Android, you navigate between activities through what is known as an intent.

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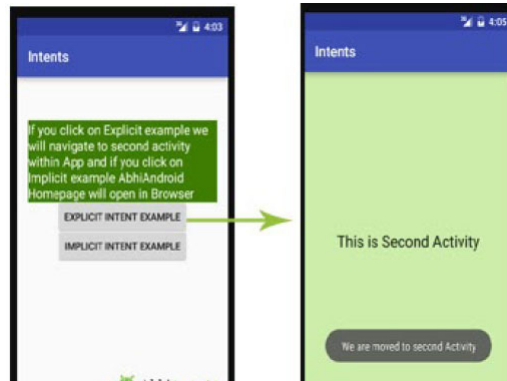
Intents in Android

- Android uses Intent for communicating between the components of an Application and also from one application to another application.

```
Intent intent = new Intent(getApplicationContext(), SecondActivity.class);  
startActivity(intent);
```

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Two Activities (two screens)



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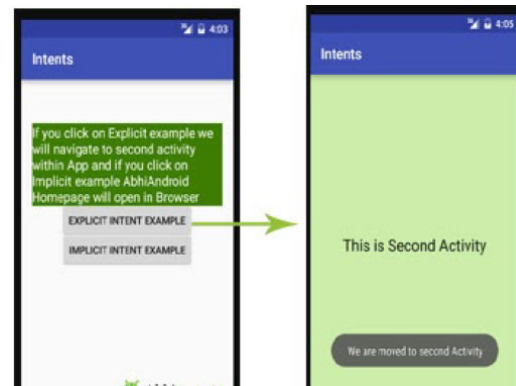
Types of Intents

- Explicit Intent: Explicit Intents are used to connect the application internally.
- Implicit Intent: In Implicit Intents we do not need to specify the name of the component. We just specify the Action which has to be performed and this action is handled by the component of another application.

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Explicit Intent

```
Intent intent = new Intent(getApplicationContext(), SecondActivity.class);
startActivity(intent);
```



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Implicit Intent

Basic example of implicit Intent is to open any web page

```
Intent intentObj = new Intent( Intent.ACTION_VIEW );
intentObj.setData( Uri.parse(" http://www.ukzn.ac.za" ) );
startActivity(intentObj) ;
```

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How it Works

```
package com.example.android.intents;
import android.content.Intent;
import android.net.Uri;
import android.support.v7.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
public class MainActivity extends AppCompatActivity {
    Button explicit_btn, implicit_btn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        explicit_btn = (Button) findViewById(R.id.explicit_intent);
        implicit_btn = (Button) findViewById(R.id.implicit_intent);

        //implement OnClick event for Explicit Intent
        explicit_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(getApplicationContext(), SecondActivity.class);
                startActivity(intent);
            }
        });

        //implement OnClick event for Implicit Intent
        implicit_btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(Intent.ACTION_VIEW);
                intent.setData(Uri.parse("http://www.ukzn.ac.za.com"));
                startActivity(intent);
            }
        });
    }
}
```

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Explicit Intent

```
explicit_btn = (Button) findViewById(R.id.explicit_intent);
implicit_btn = (Button) findViewById(R.id.implicit_intent);

//implement OnClick event for Explicit Intent
explicit_btn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Intent intent = new Intent(getApplicationContext(),
        SecondActivity.class);
        startActivity(intent);
    }
});
```

Implicit Intent

```
//implement OnClick event for Implicit Intent
implicit_btn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        Intent intent = new Intent(Intent.ACTION_VIEW);
        intent.setData(Uri.parse("http://www.ukzn.ac.za.com"));
        startActivity(intent);
    }
});
```

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Intent Uses In Android

- Android uses Intents for facilitating communication between its components like
- Activities – to start a new activity- need to pass an Intent object to startActivity() method
- Services – work in background of the application -does not require any user interface
- Broadcast Receivers – Android initiates some broadcast message on events, e.g. System reboot, low battery warning etc..